

Final Project

Due: 11 PM, Tuesday, May 19, 2026

A Unimodal Foundation Model for DESI Spectra

Overview

The goal of this project is to build a **unimodal foundation model** for astrophysical spectra. Rather than chasing the breadth of the AION-1 paper — which combined 39 different modalities (heterogeneous DESI and Subaru imaging, DESI spectra, scalar metadata such as per-band magnitudes, and so on) into a single transformer trained on 200 million observations of stars, galaxies, and quasars — your model will do one thing well: understand DESI spectra, with redshift as the only auxiliary signal.

The motivating critique of AION-1 is that, despite the ambition of its title, it does not behave like a foundation model in the meaningful sense. A foundation model should appear to *understand* its domain — generalizing to genuinely out-of-sample inputs — rather than merely finding more efficient ways to do narrow tasks (which a CNN can already do). Spreading a fixed parameter budget across 39 modalities likely prevented AION-1 from developing deep competence in any single one. Your job is to invert that trade-off: one modality, treated seriously.

The other critique concerns how AION-1 handles redshift, and addressing it is the core technical contribution you are expected to make.

Background

A DESI spectrum is a measurement of intensity as a function of wavelength for an astrophysical object. Different wavelengths carry different information: emission and absorption features at known laboratory wavelengths reveal which chemical elements are present, and the *shift* of those features from their laboratory values reveals how far away the object is, through cosmological expansion. You can browse the imaging and (where available) overlaid spectra interactively at the [Legacy Survey Viewer](#), which is the tool used in lecture to illustrate redshift measurement on real galaxies. We will do more practice of how to use the publicly available DR1 dataset on the sky viewer.

Concretely, the redshift z is defined by

$$\frac{\lambda_{\text{observed}}}{\lambda_{\text{emitted}}} = 1 + z$$

A higher z means a more distant object. For a nearby galaxy with $z \approx 0.005$, the observed wavelength is essentially the laboratory value; for galaxies billions of light-years away, z can be arbitrarily high (though < 1100 , for reasons I will not get into here). Identifying spectral features such as H α (6563 Å) or the Ca H&K doublet (~3945 Å) and measuring their observed wavelengths gives you z directly. This is what the DESI pipeline does algorithmically, and what your model must learn to do from raw spectra.

A DESI spectrum carries flux measurements on a fixed **7,081-pixel wavelength grid** spanning roughly 3,600–9,800 Å. AION-1 compresses each spectrum down to **273 learned tokens** via a separate machine-learning tokenizer (not human-engineered tokens, as in GPT-2), then appends the redshift as a **single additional scalar token** and trains a transformer encoder-decoder with masked-token prediction over all modalities jointly.

What's Wrong with the AION-1 Redshift Treatment

This is the central technical issue your project addresses.

AION-1 treats the redshift token like any of the other 273 spectral tokens: at training time, masking is applied randomly across all 274 positions, so the redshift is masked only occasionally. The redshift never makes it cleanly into the encoder's representation space, so the authors instead train a separate lightweight prediction head on top of a *frozen* encoder to recover redshift as a downstream task.

Two problems follow from this design:

1. **The redshift is not weighted appropriately.** A spectrum is mostly noise; only a few features carry real information, and that information is distributed across hundreds of tokens. The redshift, by contrast, is a single highly-distilled scalar — losing it is much more costly than losing any one spectral token. Treating them with equal weight under uniform random masking under-trains the model on the quantity that matters most.
2. **The redshift never enters the representation space.** Because the prediction head is bolted on after the encoder is frozen, the encoder's internal representation is never shaped by the requirement to know z . The redshift becomes a downstream attachment, not an organizing principle of the representation.

The AION-1 paper does not report how the model performs at predicting redshift directly — the strong indication being that the result was not impressive enough to publish.

Project Specification

Data

- **DESI Data Release 1 spectra** (specifically, the Early Data Release / SV3 "one-percent" survey — roughly 1 million stars, galaxies, and quasars), accessed through the [Multimodal Universe](#) dataset, which packages MMU-ready cross-survey data. Even at 1% of full DESI, the data is large; **start with 1% of that** (i.e., ~0.01% of the full DESI corpus) to get the training pipeline working before scaling up.
- **No imaging modalities.** No Subaru / HSC. No Legacy Survey grz photometry. No magnitudes. Spectra and redshifts only.

Modalities

- **Two:** the DESI spectrum itself, plus the redshift z . Strictly unimodal in the spectral sense; redshift is the auxiliary scalar the model must learn to predict and represent.

Architecture

- A transformer trained with masked-token prediction, in the spirit of AION-1 but with the redshift handling redesigned. You may follow the AION-1 tokenization scheme (~273 learned tokens per spectrum) or propose your own; either way, you must be able to **explain in your own words how the tokenization works**, because this will be discussed in conversation with the instructor and the TA.
- Targeting the **300M-parameter scale** is the realistic goal. AION-1 publishes variants up to 3.1B parameters; matching even the smallest variant successfully would be a strong result.

The Redshift Mechanism (Choose At Least One)

You are expected to do better than AION-1 on redshift handling. Two recommended approaches, either or both of which is acceptable:

Approach A — Joint training with a lightweight predictor. Attach a small MLP head to the representation space that predicts z , and train it *jointly* with the masked-token objective rather than after freezing the encoder. The encoder's weights are then shaped by the redshift loss, forcing the representation space to encode z .

Approach B — Always-mask the redshift token. Whatever masking strategy you use for spectral tokens, force the redshift token to be **masked on every training example**. The model is then required to reconstruct z from the spectral context every step, instead of seeing it most of the time and being asked to predict it occasionally.

A combined approach is welcome. Other principled designs are welcome too — argue for them.

Scope Limits

- No image tokens. No magnitude tokens. No Subaru. The unimodal restriction is the simplification that makes this project tractable in a course timeframe; do not break it.
- A pure redshift regressor (e.g., a CNN trained directly on spectra to output z) is **not** a foundation model and is not what is being asked for, even though it might score well on the redshift benchmark.

Evaluation

Your model will be tested on a **held-out benchmark set that you do not have access to during development**, prepared by the instructor. The evaluation has two components:

1. **Redshift prediction.** Given an unseen spectrum, predict z . Accuracy will be compared against the DESI pipeline values.
2. **Spectrum reconstruction.** Given a partially masked spectrum the model has never seen, reconstruct the missing portions.

The benchmark set will include **spectra not taken by DESI** — out-of-distribution data from a different instrument. Generalizing to these is the test of whether the model has learned something foundational about spectra, or merely something instrument-specific about DESI. Strong out-of-sample performance is what earns the "foundation model" label.

Tools and Resources

- **Data:** [Multimodal Universe](#) (the DESI Early Data Release / 1% subset is the relevant slice). Be aware of storage constraints; staying spectral-only keeps the footprint manageable.
- **Reference implementation:** the [AION-1 GitHub repo](#) is public — code, tokenizers, and evaluation suite are released open-source. You may study it and feed it to Gemini, Claude, or other LLMs to help understand it — but **you are not reproducing it**. You are doing something deliberately different (unimodal, with a redesigned redshift mechanism).
- **Pretrained checkpoints:** AION-1 weights are on Hugging Face under the [polymathic-ai](#) org. The lightest variant — [aion-base](#) (300M parameters) — is the most accessible starting point; the larger 800M and 3.1B variants were released within the past week or so as of the lecture.

- **LLM and agentic-coding assistance:** explicitly encouraged for implementation work. The benchmark is the test, so use whatever helps you get there.

Deliverables

1. **A trained model** that ingests DESI spectra and produces (a) redshift predictions and (b) reconstructions of masked spectral regions.
2. **Benchmark results** on the held-out instructor set, including performance on out-of-distribution (non-DESI) spectra.

Why This Project

The long-term vision motivating this assignment is an astrophysics foundation model that eventually understands more astrophysics than any single human expert — the AlphaGo trajectory, but for a scientific domain. AION-1 is a pioneering first attempt and deserves credit for the breadth of data it integrated, but it does not yet behave like a foundation model in the way the instructor means. By restricting scope to spectra and getting the redshift mechanism right, this project is a step toward what such a model could actually look like.

For More Background

New York Times article on discoveries made by DESI on March 19, 2025:

<https://www.nytimes.com/2025/03/19/science/space/astronomer-desi-dark-energy.html>